

PRAKTIKUM SISTEM OPERASI

MODUL 15

Keamanan Linux



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. 1. Integritas: dasar hashing [10 Point]

- a. [2 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file /etc/passwd. Berapa nilai hash dari file /etc/passwd? Screenshot nilai hash dari file tersebut.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum /etc/passwd
4fffd7c88778adc3eee8dfb91ba69c1e3845933a86c290d56e9ed4ab22ab250 /etc/passwd
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ shasum512 /etc/passwd
shasum512: command not found
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum /etc/passwd
2cca4fcdde5f2c7f384182c565a3e1c23635b037f87ea58fbc53f66d81404009f8bbf58c389e663
8cd03c6699893950093c1e005a3590ab52b7924988aa2cca5 /etc/passwd
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum /etc/passwd
806e70cbc0edaa5ed65936e1a5b32802 /etc/passwd
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

- b. [2 Point] Buatlah file bernama test_0.txt pada folder /home/praktikan. Isi file tersebut isi yang ada di file /etc/passwd (copy paste isi file /etc/passwd ke test_0.txt)

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ cp /etc/passwd test_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ cat test_0.txt
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:/:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sb
```

- c. [2 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_0.txt. Berapa nilai hash dari file test_0.txt? Screenshot nilai hash dari file test_0.txt.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum test_0.txt
4fffd7c88778adc3eee8dfef91ba69c1e3845933a86c290d56e9ed4ab22ab250 test_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum test_0.txt
2cca4fcdde5f2c7f384182c565a3e1c23635b037f87ea58fbc53f66d81404009f8bbf58c389e663
8cd03c6699893950093c1e005a3590ab52b7924988aa2cca5 test_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum test_0.txt
806e70cbc0edaa5ed65936e1a5b32802 test_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

- d. [2 Point] Rename file test_0.txt menjadi file_0.txt. Lakukan hash SHA256, SHA512 dan MD5 untuk file_0.txt. Berapa nilai hash file_0.txt? Screenshot nilai hash dari file_0.txt.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mv test_0.txt file_o.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mv test_0.txt file_0.txt
mv: cannot stat 'test_0.txt': No such file or directory
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mv file_o.txt file_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum file_0.txt
4fffd7c88778adc3eee8dfef91ba69c1e3845933a86c290d56e9ed4ab22ab250 file_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum file_0.txt
2cca4fcdde5f2c7f384182c565a3e1c23635b037f87ea58fbc53f66d81404009f8bbf58c389e663
8cd03c6699893950093c1e005a3590ab52b7924988aa2cca5 file_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum file_0.txt
806e70cbc0edaa5ed65936e1a5b32802 file_0.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

- e. [2 Point] Apa hasil pengamatan Anda? File apa saja yang mempunyai hash yang sama? Jelaskan!

File /etc/passwd, test_0.txt, dan file_0.txt memiliki nilai hash yang sama karena isi file tidak berubah. Perubahan nama file tidak memengaruhi nilai hash, karena hash dihitung berdasarkan isi file, bukan nama file.

2. Integritas: avalance [15 Point]

2.

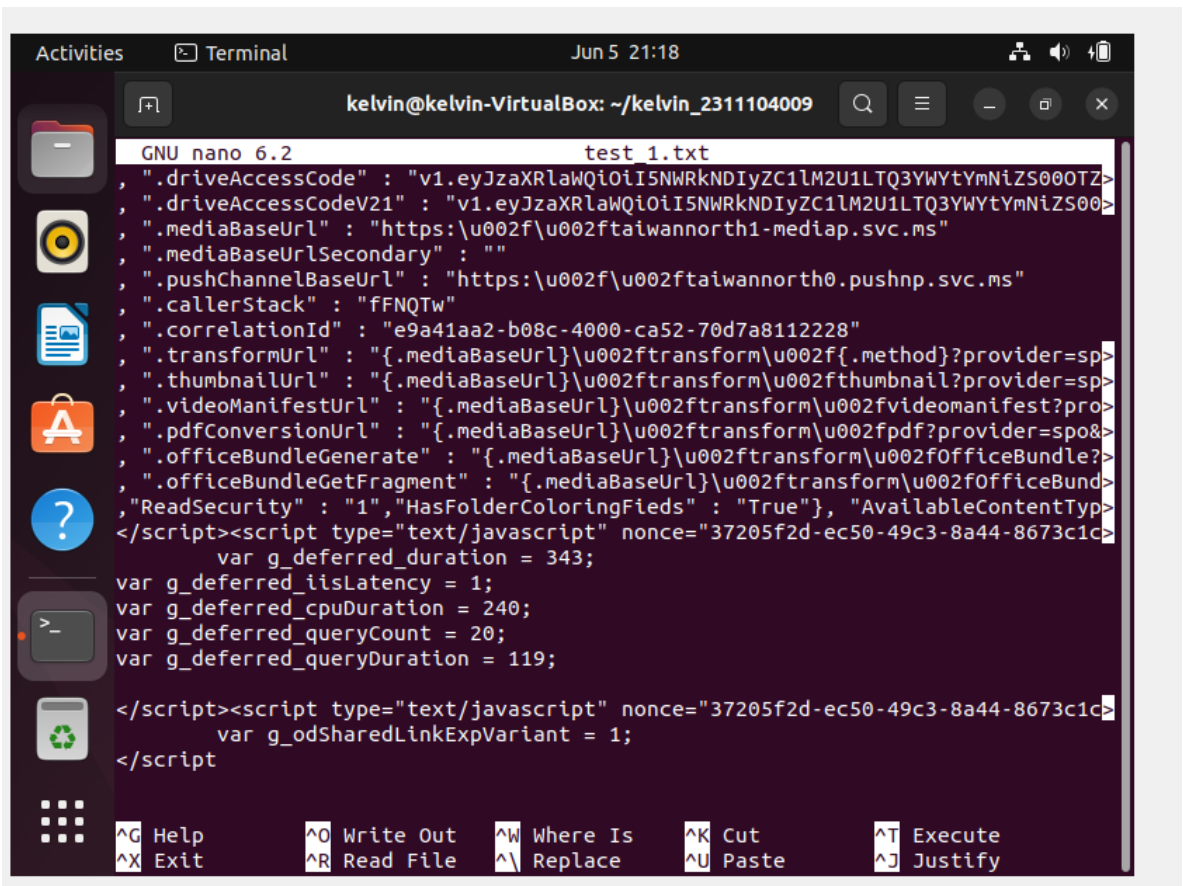
- a. [3 Point] Download file bernama test_1.txt di link ini tiny.cc/test1_txt

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ wget http://tiny.cc/test1_txt -O
test_1.txt
--2026-06-05 21:13:25-- http://tiny.cc/test1_txt
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)[157.245.113.153]:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://tiny.cc/test1_txt [following]
```

- b. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.txt. Berapa nilai hash test_1.txt? Screenshot nilai hash dari file test_1.txt.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum test_1.txt
b7ec7a8109c88e9cf05bd80f7920d6ffd720a79ce39afaf55f59afb769f702da  test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum test_1.txt
c955fab08275f1d537a369add4f065298d903220e65252b4ad268826c5cbcbd8be90ab61579e68f
348b863a11e30f75f534bff53abd8cd141737d3ecf83463a0  test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum test_1.txt
d32cbeca5c95dceccdb79e8374cce3d  test_1.txt
```

- c. [3 Point] Hapuslah titik diakhir file test_1.txt tersebut, simpan file tersebut!



- d. [3 Point] Lakukan hash dari SHA256, SHA512 dan MD5. Screenshot nilai hash dari file test_1.txt.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum test_1.txt
f22fbc320657905662fa149001a5dcfc4ff99a59fbf8e827cfff1769cdc3847d2  test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum test_1.txt
2073b8973f0a5db6f952f72c7a6f69fcede107bc29c6e10be9ac5247b76da485a643077a6554361
2d04baec3d4a3e790c64fb75247a047c5d1260c7bc2a6aeca  test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum test_1.txt
360116b81953834c583c617d757585f9  test_1.txt
```

- e. [3 Point] Apa analisis (hasil pengamatan) Anda mengenai hal tersebut! Apakah nilai hash sama?

Nilai hash berubah total meskipun hanya satu karakter yang dihapus. Hal ini menunjukkan sifat avalanche effect, yaitu perubahan kecil pada isi file akan menghasilkan nilai hash yang sangat berbeda.

3. Integritas: metadata [18 Point]

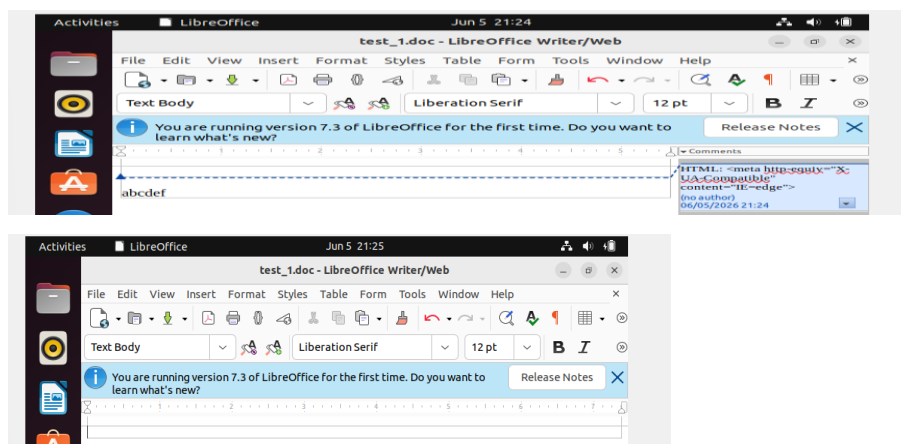
- a. [3 Point] Download file bernama test_1.doc di link ini tiny.cc/test1_doc

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ wget http://tiny.cc/test1_doc -O test_1.doc
4URL transformed to HTTPS due to an HSTS policy
--2026-06-05 21:21:49-- https://tiny.cc/test1_doc
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversitiofficial.my.chasepoint.com/wwp/a/personal/inf
```

- b. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.doc. Screenshot nilai hash dari file test_1.doc.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum test_1.doc
849d8a9679850824a0d29880549df50850dc6d50a2d1ca71c84e5183ac9a3ddd test_1.doc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum test_1.doc
2d6e42eba30baa1a369310d2f1da422e674eb488c2ae0a88659dbfb3f3bdc5c51919ba636305c3
80501064203530122890d557632accf74438da18bf3da11a4 test_1.doc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum test_1.doc
17321ad238d52bdf5030aa1ec8d42c3e test_1.doc
```

- c. Buka kembali file test_1.doc. Lakukan hal ini:
- [3 Point] Ketik abcdef. Save test_1.doc
 - [3 Point] Hapus abcdef. Save test_1.doc



- d. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.doc. Screenshot nilai hash dari file test_1.doc.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha256sum test_1.doc
f17b9a81959140cab1da1ce112e7f24a56c57b6f7f4731870a54dc1e1a0faeae test_1.doc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ sha512sum test_1.doc
b9224dd5a2b087391e95a6bc4b236f68b64b88a98f78ad87d22787910c542e416ddeac2652db8b1
b423e4977fec9166bf80292359c0767c0460e22faabe0662 test_1.doc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ md5sum test_1.doc
7c2154cabf772bbcd18a593554826b22 test_1.doc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

- e. [3 Point] Hasil pengamatan apa yang diperoleh? Jelaskan alasannya!

Nilai hash berubah walaupun isi dokumen dikembalikan seperti semula. Hal ini terjadi karena file dokumen menyimpan metadata, seperti waktu penyimpanan, riwayat perubahan, atau informasi aplikasi. Jadi, perubahan metadata juga memengaruhi nilai hash.

4. **Konfidensialitas: encfs [27 Point]**

- a. [5 Point] Jalankan perintah berikut:

```
encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_normal
```

Pilih y (enter), pilih y (enter), enter, kemudian buatlah password. **Ingatlah password yang dibuat.** Setelah selesai, perhatikan bahwa telah muncul folder bernama folder_normal dan folder terenkripsi.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mkdir -p ~/Kelvin_2311104009/folder_terenkripsi
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mkdir -p ~/Kelvin_2311104009/folder_normal
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ encfs ~/Kelvin_2311104009/folder_terenkripsi ~/Kelvin_2311104009/folder_normal
Creating new encrypted volume.
Please choose from one of the following options:
  enter "x" for expert configuration mode,
  enter "p" for pre-configured paranoia mode,
  anything else, or an empty line will select standard mode.
?> y

Standard configuration selected.

Configuration finished. The filesystem to be created has
the following properties:
```

- b. [5 Point] Copy dua atau tiga buah file (file apa saja) ke folder_normal. Amati dan tulis hasil observasi Anda pada folder_terenkripsi!

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ echo "file pertama" > folder_normal1/file1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_acak
Qn1KI8-ZRfL1MaXVCZPp7ToB
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ echo "file kedua" > folder_normal1/file2.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ echo "file ketiga" > folder_normal1/file3.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_normal1
file1.txt file2.txt file3.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_acak
0UqGBw-Fw6y,con1FOXc2b6y 63dmrneCdKWIUyOSZRxiPDjg Qn1KI8-ZRfL1MaXVCZPp7ToB
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

File yang dimasukkan ke folder_normal muncul di folder_terenkripsi dalam bentuk nama dan isi yang terenkripsi, sehingga tidak dapat dibaca secara langsung.

- c. [5 Point] Hapus salah satu file (bebas) pada folder_terenkripsi. Amati dan tulis hasil observasi Anda pada folder_normal!

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_acak
0UqGBw-Fw6y,con1FOXc2b6y 63dmrneCdKWIUyOSZRxiPDjg Qn1KI8-ZRfL1MaXVCZPp7ToB
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ rm folder_acak/^C
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_acak/Qn1KI8-ZRfL1MaXVCZPp7ToB
folder_acak/Qn1KI8-ZRfL1MaXVCZPp7ToB
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_normal1
file1.txt file2.txt file3.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_acak/Qn1KI8-ZRfL1MaXVCZPp7ToB
folder_acak/Qn1KI8-ZRfL1MaXVCZPp7ToB
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_normal1
file1.txt file2.txt file3.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ rm folder_acak/63dmrneCdKWIUyOSZRxiPDjg
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_normal1
file1.txt file3.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

Ketika salah satu file terenkripsi dihapus, file aslinya pada folder_normal juga hilang karena kedua folder saling terhubung.

- d. [5 Point] Lakukan umount dengan perintah:

```
Fusermount -u ~/folder_anda/folder_normal
```

Amati dan tulis hasil observasi Anda pada folder_normal dan folder_terenkripsi

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ fusermount -u ~/kelvin_2311104009/folder_normal1
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_normal1
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_acak
0UqGBw-Fw6y,con1FOXc2b6y Qn1KI8-ZRfL1MaXVCZPp7ToB
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

Setelah di-unmount, folder_normal menjadi kosong/tidak menampilkan file asli, sedangkan folder_terenkripsi tetap berisi file terenkripsi.

- e. **[7 Point]** Buatlah folder baru bernama folder_sembarang. Lakukan perintah berikut ini:

```
encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_sembarang
```

Amati dan tulis hasil observasi Anda!

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mkdir folder_pipin
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ encfs ~/kelvin_2311104009/folder_Acak ~/kelvin_2311104009/folder_pipin
The directory "/home/kelvin/kelvin_2311104009/folder_Acak/" does not exist. Should it be created? (y,N) y
Creating new encrypted volume.
Please choose from one of the following options:
  enter "x" for expert configuration mode,
  enter "p" for pre-configured paranoia mode,
  anything else, or an empty line will select standard mode.
?> ^C
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ encfs ~/kelvin_2311104009/folder_acak ~/kelvin_2311104009/folder_pipin
EncFS Password:
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls folder_pipin
file1.txt  file3.txt
```

File asli dapat muncul kembali di folder_sembarang setelah folder terenkripsi di-mount menggunakan password yang benar.

5. **Konfidensialitas: gpg [30 Point]**

- a. **[5 Point]** Membuat kunci publik dan privat. Jalankan perintah ini:

```
gpg --gen-key
```

Baca dan ikuti perintah yang ada dari program gpg!


```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --gen-key
gpg (GnuPG) 2.2.27; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Note: Use "gpg --full-generate-key" for a full featured key generation dialog.

GnuPG needs to construct a user ID to identify your key.

Real name: kelvin
Email address: kelvinferdinan24@gmail.com
You selected this USER-ID:
    "kelvin <kelvinferdinan24@gmail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key C76E690E0A1E9C38 marked as ultimately trusted
gpg: directory '/home/kelvin/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/kelvin/.gnupg/openpgp-revocs.d/A1B
B5F3F9BE07C718AADC56BC76E690E0A1E9C38.rev'
public and secret key created and signed.
```

- b. **[5 Point]** Hasil publickey dan privatekey ada di folder ~/.gnupg. Privatekey tidak boleh keluar dari komputer ini dan hanya Anda saja yang dapat mengaksesnya. Kunci publik akan dibagikan kepada seluruh dunia atau untuk orang yang Anda inginkan saja. Kunci publik adalah pubring.gpg dan kunci private adalah secring.gpg
Lakukan perintah berikut ini untuk mengetahui daftar key yang Anda punya dan fingerprint kunci yang Anda punya!

```
gpg --list-keys
gpg --fingerprint nama@email.com
```

Screenshot hasil perintah di atas!

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-04
/home/kelvin/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
       A1BB5F3F9BE07C718AAD56BC76E690E0A1E9C38
uid           [ultimate] kelvin <kelvinferdinan24@gmail.com>
sub   rsa3072 2026-06-05 [E] [expires: 2028-06-04]

kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --fingerprint kelvinferdinan24@gmail.com
pub   rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
       A1BB 5F3F 9BE0 7C71 8AAD C56B C76E 690E 0A1E 9C38
uid           [ultimate] kelvin <kelvinferdinan24@gmail.com>
sub   rsa3072 2026-06-05 [E] [expires: 2028-06-04]
```

- c. [5 Point] Export kunci publik Anda. Jalankan perintah ini:

```
gpg --armor --export nama@email.anda > mypublic_key.asc
```

File **mypublic_key.asc** adalah file yang akan dibagikan kepada teman-teman Anda. Teman Anda akan menggunakan kunci publik Anda jika ingin mengirim pesan rahasia kepada Anda. Hanya Anda yang dapat membaca pesan tersebut karena hanya mempunyai kunci privat yang bersesuaian dengan kunci publik Anda.

- Rename **mypublic_key.asc** menjadi **nim_anda.asc**, Contoh: 130118xxxx.asc
- Taruh file **nim_anda.asc** ke folder pada link berikut: <http://tiny.cc/SisopNomor5C>

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --armor --export kelvinferdinan24@gmail.com > mypublic_key.asc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mv mypublic_key.asc 2311104009.asc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls
2311104009.asc  folder_acak  folder_normal  folder_pipin  test_1.doc
file_0.txt     folder_Acak  folder_normal1 folder_terenkripsi test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ cat 2311104009.asc
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQGNBGoI7ysBDADahFagcLvl2b1jsewWYyeNw2mSTF9NqLXnAKxfcvMpFv9+fTy/
lnAihQUAO6opP4KbIwQhwLvLnJrsLu8a30IjTz3r5vFEERun/w0vhi2omiYik9pU
QrzvuYHcs9jnXYj4fNT3bIs6bxf5hqmTRn4uS0f1tUaIVNm6bvqJxLA1H9N0xA2z
OUePC+2SwzZNemNj9S9nmXyitiUeAaC6lRS0/RyX3DE13pGBUBgGJ8vitIpJvUfj
a17+KR3BZ+qc37tV671QCK3winsIXtZnLPxunWdyRi3ACsXN1B3r8bQJNHkLV02
ZRRQyvVlVwDW/LnF5TBAZmkPY3gBuhEFy9qE94E0jJmL48lY5jlaPm9E5jjQ7Ktp
2EBtfsB5PvHjs+KXRIKbIFxeKKEJdyW+nwIhskATqw1JXVmSPQLGC0+MLAqSSd
lRRh8+n7l72cLgW4YzQ6l0x1IKXoufzg6qF0x2Xe0T+ELZ14DNgcOKXvXMcAPMC6
sfN60YkjPrZBb5EAEQEAAABQja2VsdmluIDxrZWx2aW5mZXJkaW5hbGJlOQGdtYWls
LmNvbT6JAdQEEwEKAD4WIQShu18/m+B8cYqtxWvHbmKOCh6cOAUCaILvKwIbAwUJ
A8JnAAULCQGHAgYVCgkICwIEFgIDAQIeAQIXgAAKCRDHbmkOCh6cOKm0DAC6iFEC
L+aiSdssJYwean9nYRvFu4fH53Ejvmy2oggcfKCQgjqNsly/u1QA+703qst/qKBq
o27tfQZVE3F19tZspkMTGd1L5Rq2KLp2mHa8UaBkKR420N0+aYeg+9z4sST+bteS
mmCKMlyCug0Zb8OLKusdhwZmWntYVpsq1R3iJwDL4hX2Cb5DG6IiULub0jlRuMXt
ihJCylot+mCDNjnZ4XbSzahWh43My280ebW64FZCkehj1k1JvnEIsydsNhd+GuJ9
3C4TbP1S+HL9HUFSPpZplbWfQLPt4WyCwrME5gS8e5tEWJfQx9ecEINwgcNoW25X
IeXtunk5GDAoR8xxRTRprT9fQGE7nWief3x8Z9n0gIoS7cxaJYy8CLAgQ/rGob3
vfoamYzUHM9gmYUyZwqQLjy1rog537tQL3zU0gI+V8Rv/ekWmm7S6aVEgMmR8yL
SauMh3Kl+L3qCCB2Qb8lOmFYQ5ShaXusv+u67UZ2cg8Kru8Fr2+FsIwep/Fq5AY0E
```

- d. [5 Point] Mengimport kunci publik orang lain. Silakan download file `nim_teman_sebelah_anda.asc` dan jalankan perintah ini untuk mengimport (menambahkan kunci publik orang lain ke sistem Anda):

```
gpg --import nim_teman_sebelah_anda.asc
```

Contoh: “`gpg --import 130118yyyyy.asc`” Setelah mengimport kunci publik teman Anda, Anda dapat mengirimkan pesan kepada teman Anda secara terenkripsi.

Untuk memastikan bahwa kunci publik telah diimport lakukan perintah

```
gpg --list-keys
```

dan nama teman Anda ada pada hasil perintah tersebut.

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --import 2311104009.asc
gpg: key C76E690E0A1E9C38: "kelvin <kelvinferdinan24@gmail.com>" not changed
gpg: Total number processed: 1
gpg:      unchanged: 1
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid: 2  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 2u
gpg: next trustdb check due at 2028-06-04
/home/kelvin/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
      A1BB5F3F9BE07C718AAD56BC76E690E0A1E9C38
uid           [ultimate] kelvin <kelvinferdinan24@gmail.com>
sub   rsa3072 2026-06-05 [E] [expires: 2028-06-04]

pub   rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
      4BA1CD423259A8ACD34BB161059CA2531E9581BB
uid           [ultimate] copia <copia@gmail.com>
sub   rsa3072 2026-06-05 [E] [expires: 2028-06-04]

kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

- e. [5 Point] Enkripsi pesan. Buatlah file bernama `file_rahasia.txt`. Isi file tersebut dengan pesan rahasia Anda. Pesan inilah yang akan Anda kirim ke teman Anda. Jalankan perintah ini untuk melakukan enkripsi:

```
//ditulis dalam satu baris
gpg --encrypt --armor -r
alamat_email_teman_anda_yang_baru_diimport@xxx.com file_rahasia.txt
```

```
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ nano file_rahasia.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg --encrypt --armor -r kelvinfe
rdinan24@gmail.com file_rahasia.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ ls
2311104009.asc  file_rahasia.txt.asc  folder_normal  folder_terenkripsi
file_0.txt      folder_acak           folder_normal1  test_1.doc
file_rahasia.txt  folder_Acak          folder_pipin    test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ mv file_rahasia.txt.asc file_raha
sia_untuk_2311104009.asc
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$
```

- f. **[5 Point]** Hanya teman anda yang dapat membuka file tersebut. Hasil dari proses tersebut adalah file_rahasia_untuk_teman_anda.asc (contoh: file_rahasia_untuk_130118yyyy.asc)
Taruh file_rahasia_untuk_teman_anda.asc ke folder pada link:

<http://tiny.cc/SisopNomor5F>

Download dan buka file yang diperuntukkan bagi Anda dan lakukan dekripsi untuk melihat isi pesan dengan perintah:

```
gpg file_rahasia_nim_anda.asc
```

```
-rw-rw-r-- 1 kelvin kelvin 353895 Jun  5 21:17 test_1.txt
kelvin@kelvin-VirtualBox:~/kelvin_2311104009$ gpg -d file_rahasia_untuk_2311104
009.asc
gpg: encrypted with 3072-bit RSA key, ID 2691A1A6D7D0A9E9, created 2026-06-05
      "kelvin <kelvinferdinan24@gmail.com>"
1234567891287127
```